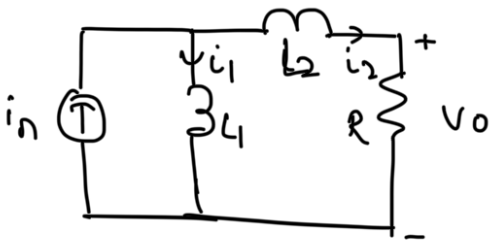
Voltage infinite ⟶ Inductor, Resistor, current source in a loop

Current infinite ⟶ Current source, Capacitor in a loop



$$L_1\frac{di_1}{dt} = L_2\frac{di_2}{dt} + i_2R$$

$\downarrow \infty$ $\downarrow \infty$ $\quad i_1, i_2 \to$ not ∞

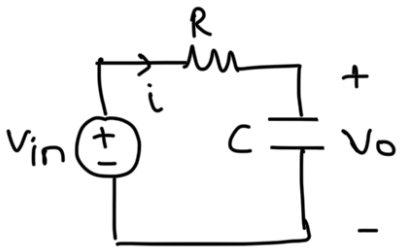
$$\Rightarrow L_1 i_1 = L_2 i_2 \qquad i_1 + i_2 = i_n$$

$$i_1(t=0^+) = \frac{L_1 I_n}{L_1+L_2} \qquad i_2(t=0^+) = \frac{L_2 I_n}{L_1+L_2}$$

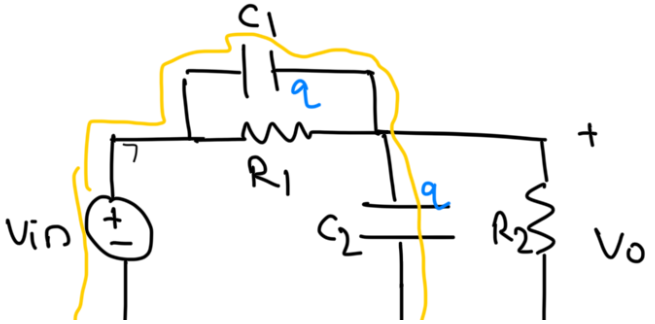
$$V_o(t=0^+) = -\frac{L_2 I_n R}{L_1+L_2} \qquad V_o(t=\infty) = 0$$

$$V_o = V_o(t=0^+)e^{-t/\tau} + V_o(t=\infty)(1-e^{-t/\tau})$$

$$\tau = (L_1+L_2)/R$$



$$V_o = V_{in}(1-e^{-t/\tau}) \qquad \tau = RC$$



there can't be infinite voltage but there can be infinite current

$V_{in} = q/C_1 + q/C_2$



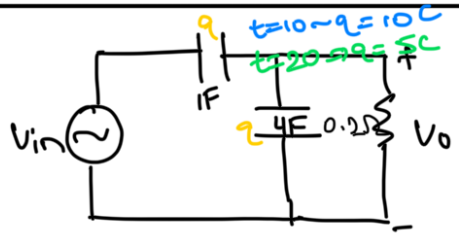
infinite current path

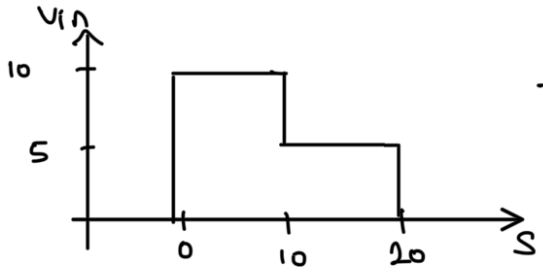
$$q = \left(\frac{C_1 C_2}{C_1 + C_2}\right) V_{in} \quad @ t = 0^+$$

$$I_1 = 1/R_1 \left(\frac{C_2 V_{in}}{C_1 + C_2}\right) \qquad I_2 = \frac{1}{R_2}\left(\frac{C_1 V_{in}}{C_1 + C_2}\right)$$

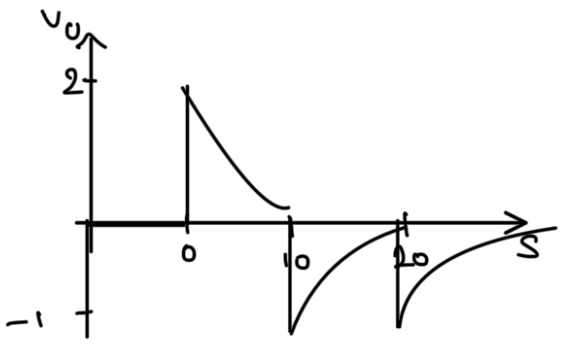
$$V_0(t=0) = I_2 R_2 = \left(\frac{C_1}{C_1 + C_2}\right) V_{in} \qquad V_0(t=\infty) = (V_{in})\left(\frac{R_2}{R_1 + R_2}\right)$$

$$\tau_{eff} = \left(\frac{R_1 R_2}{R_1 + R_2}\right)(C_1 + C_2)$$ (just remove the source)





$\tau = (1+4)(0.2) = 1$



$10 = q + q/4 \Rightarrow q = 8C$

$5 = \frac{10+q}{1} + \frac{q}{4} \Rightarrow q = -4$

$0 = \frac{5+q}{1} + \frac{q}{4} \Rightarrow q = -4$